= 1.95 × 10 V or 2.0 × 10 V

6(a)(i) As the half-life of X (in years) is very long, it means that the decay constant is very, very small. Thus the fraction of nuclei that would decay during the time of measurement is very low, and the number of nuclei N in the sample remains almost constant. [APPLICATION]

Since activity, A = λN [CONCEPT], then the activity will remain almost constant. [CONCLUSION]

(ii) **1.** Decay constant of Y, $\lambda_Y = \dfrac{\ln 2}{t_{1/2}} = \dfrac{0.693}{1.5 \times 60 \times 60}$ = 1.283 x 10^{-4} s^{-1} = 1.3 x 10^{-4} s^{-1}

2. Equilibrium is reached when the rate of production of Y (from the decay of X) is equal to its rate of decay of Y. Hence, the number of isotope Y in the sample will stabilise at a constant value.

3. Thus, the activity of Y is about 1.1 x 10^7 Bq.

Amount of Y, $N_Y = \frac{A_Y}{\lambda_Y} = \frac{1.1\times10^7}{1.283\times10^{-4}}$ = 8.6 x 10^{10} atoms

b(i) By $N = N_o e^{-\lambda t}$: [CONCEPT]

$$5N = 6Ne^{-\lambda t}$$

Take ln on both sides,

$$\ln 5 = \ln 6 - \lambda t$$

$t = \frac{\ln 6 - \ln 5}{4.95\times10^{-11}} = 3.683\times10^9$ yrs = 3.7 x 10^9 years

(ii) Decay of Th-232 will give rise to a radioactive series where there will be a number of radioactive daughter products before ending up as the stable Pb-208. It is assumed that these intermediate radioactive daughter products have very short half-lives (much shorter than that of Th-232) so the number of intermediate daughter products are insignificant compared to Th-232 and Pb-208.

(iii) If the assumption is not valid, there would have been more Th-232 in the beginning which have decayed and is still in the form of the intermediate daughter products, therefore the fraction of undecayed Th-232 is less than $\frac{5}{6}$. This means that the rock would have been older, as a longer time would have been elapsed, thus answer for **(b)(i)** will be an under-estimate.

7(a) Potential difference used to accelerate the ions, ΔV = 1060 – (–225) = 1285 V